Introduction

In Project Part 4, we'll build a prototype to evaluate our system. To do pre-testing on our system, we will develop a high-fidelity and interactive prototype with Figma. Our priority will be to deliver a functional prototype that closely matches the final implementation of the three major objectives.

The usability test consisted of three users and three tasks, and it was conducted in three different locations. Yuhin will be in charge of testing our website for user one, Evelyn for user two, and Zhiming for user three. Before conducting the interview, we developed a prototype and briefed on what was expected of us. During the interview, we read them the task requirements and requested them to execute the tasks in order to determine whether there were any design flaws in our prototype. The three tasks are: search location, search image, and search history.

The report consists of:

1. **Screenshots of the prototype**, a Figma prototype for users testing

2. **Briefing notes**, a brief information that will be provided to users before they use our system which include the purpose of our website and the required tasks.

3. **Testing with users**, the videos of our users where they are testing our website according to the instruction in briefing notes.

4. **Observations**, description about what is happening during the execution of tasks and also a summary result interviews from each task.

5. **Findings**, the problems and solutions that we have discovered from user testing through the Figma prototype.

Screenshots of prototype

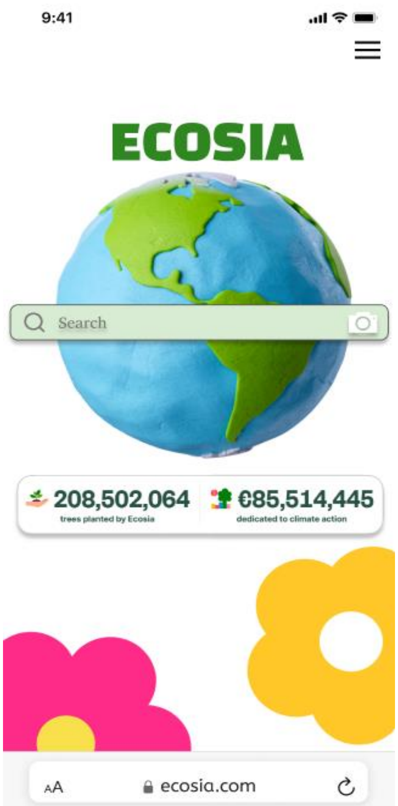


Figure 1: Search page

Figure 2: Menu

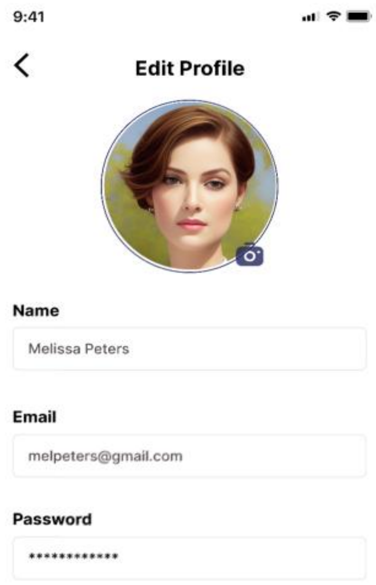


Figure3:Profile

Figure 4:Setting

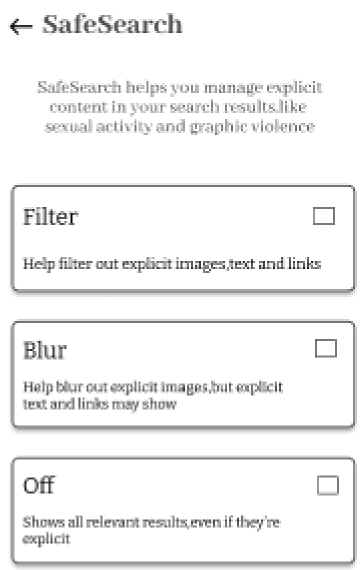


Figure 5:Safe Search

Figure 6:About us

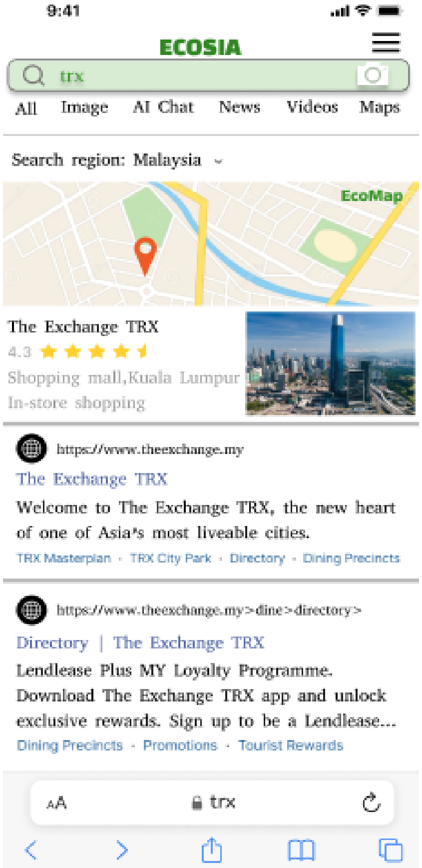


Figure 7: Result Browsing

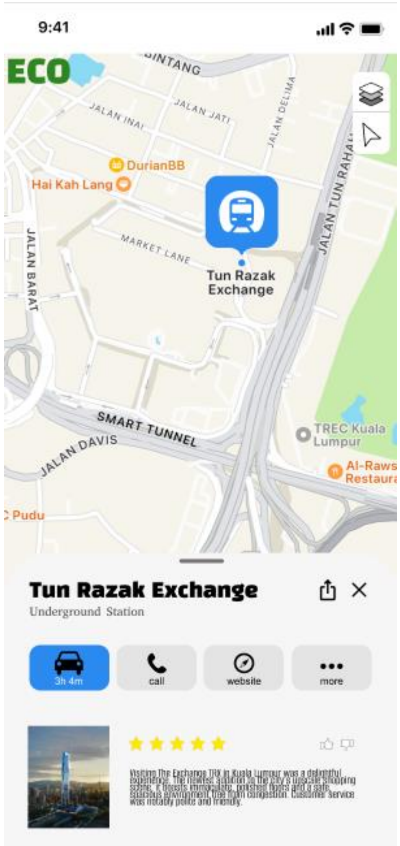
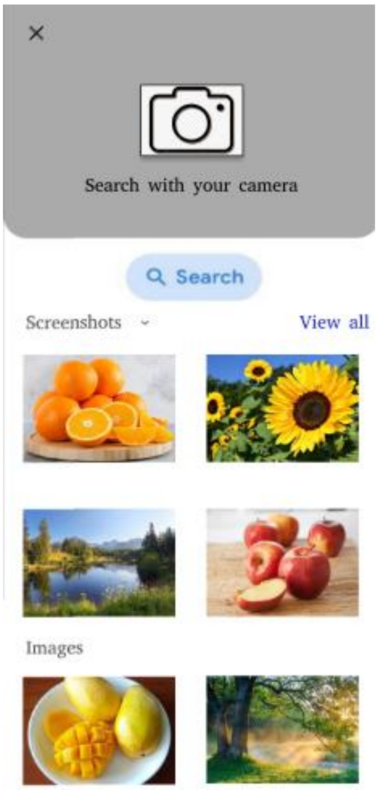


Figure 8:EcoMap



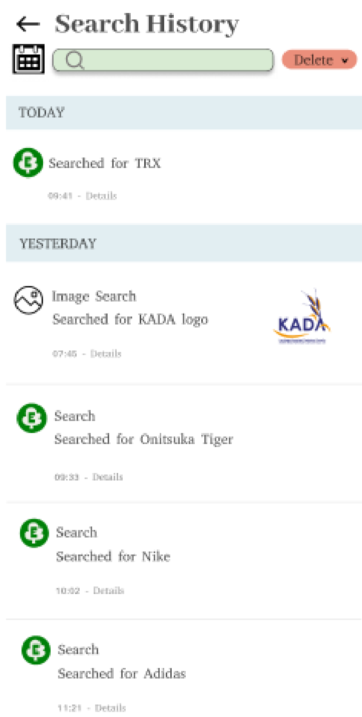


Figure 9:Search Image

Figure 10:Search History

Briefing Notes

Ecosia is a search engine dedicated to environmental sustainability by using its ad revenue to fund reforestation projects globally. Like other search engines, it enables users to search the web, but with a unique mission: for every search made, Ecosia earns money from advertisements and allocates a significant portion of this income to plant trees.

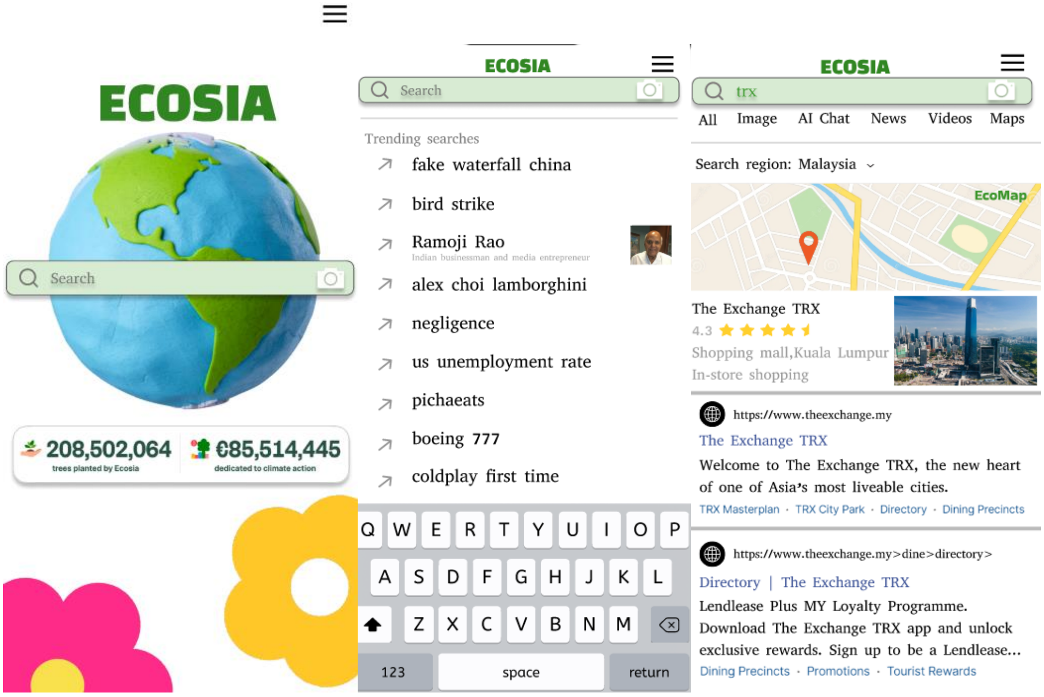
We initially chose to optimise Ecosia aim to continue promoting its admirable mission and to address some design and internal application flaws we identified in the original search engine. By enhancing the current platform, we've introduced an improved version of the Ecosia search engine, making it more user-friendly and accessible to a broader audience. This improvement not only enhances user experience but also supports our sustainability goal of "Life on Land," contributing to global reforestation efforts.

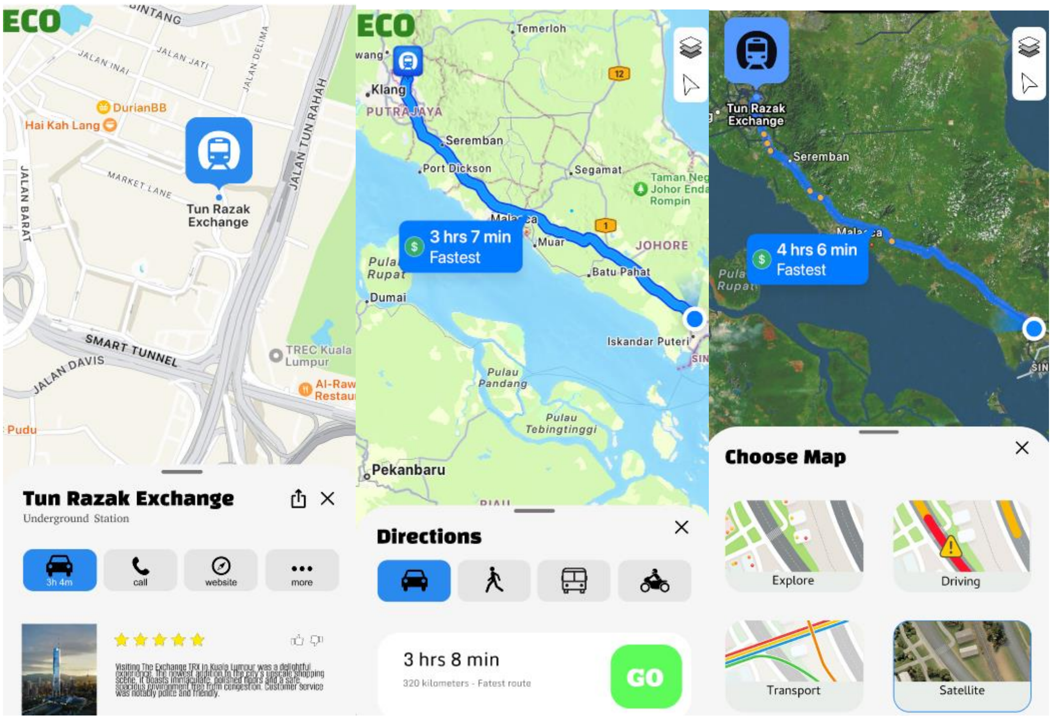
The 3 tasks that you, our user, will be testing by navigating through our paper prototype is as below:

Task 1: Use the **satellite** layer to find directions to TRX (drive)
Task 2: Use image searching to capture an apple in **HD** and find the related content
Task 3: Use history searching feature to check browsed content on 4th June by using calendar feature and **clear** the history of 3rd June.

Testing with Users

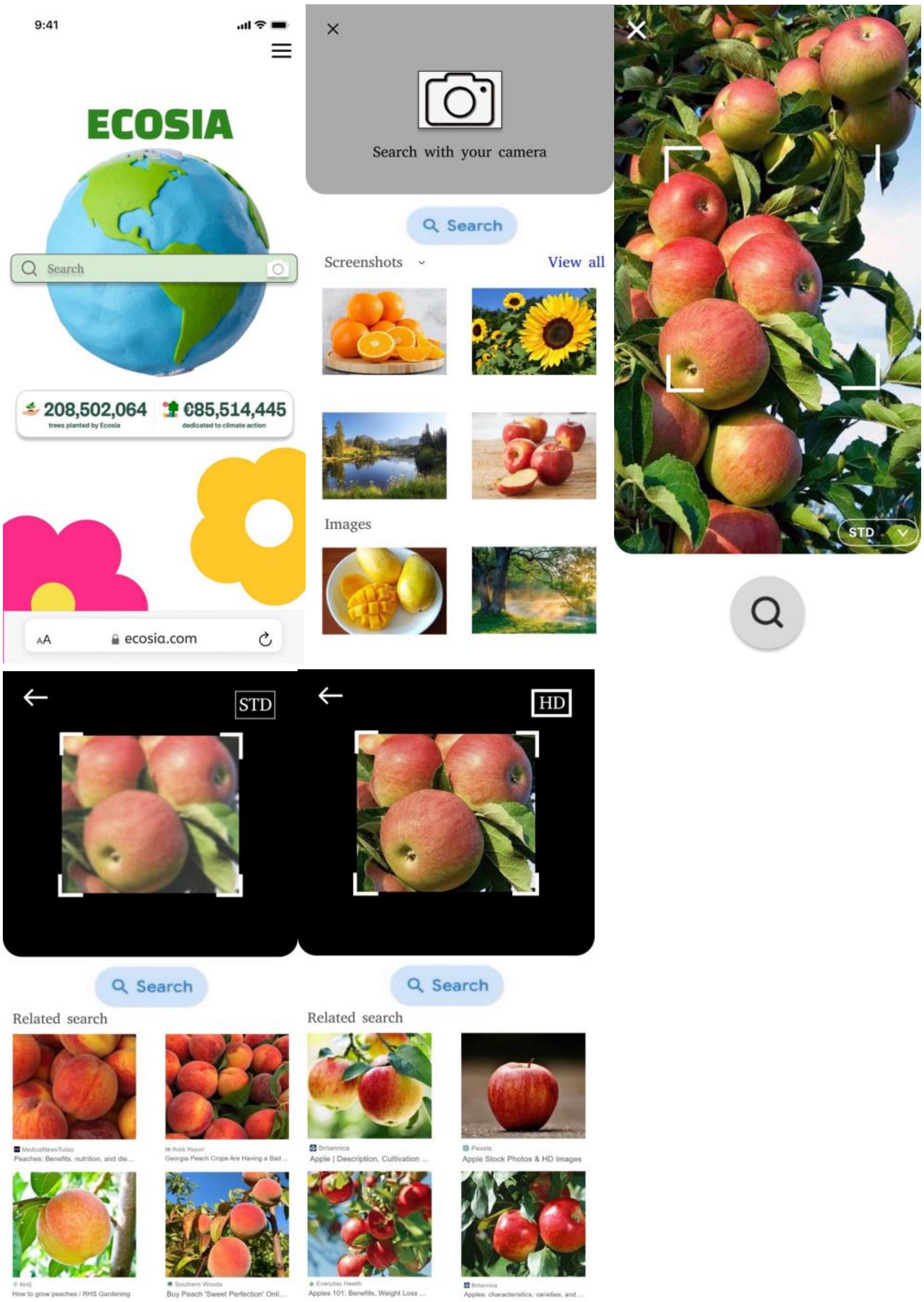
Task 1: Use the **satellite** layer to find directions to TRX (drive)





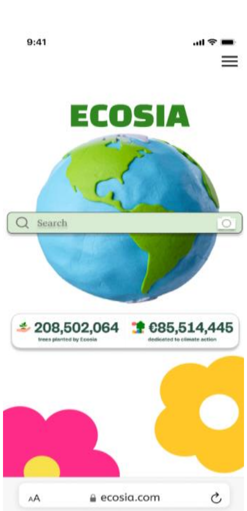
User 1: https://youtu.be/BTCJzGNONEU
User 2: https://youtu.be/UBQoZi9eDW8
User 3: https://youtu.be/M5hc6YfwmXo

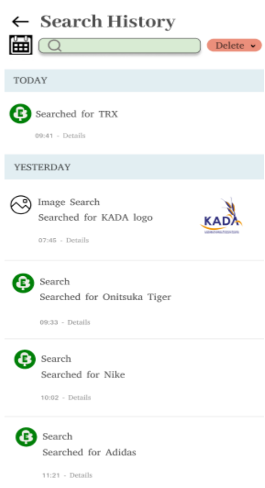
Task 2: Use image searching to capture an apple in **HD** and find the related content

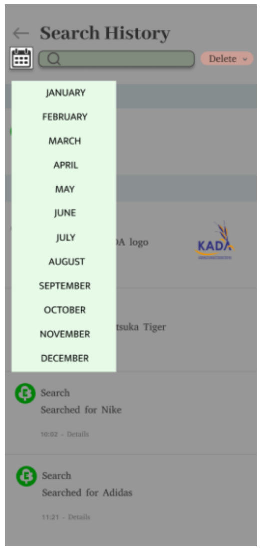


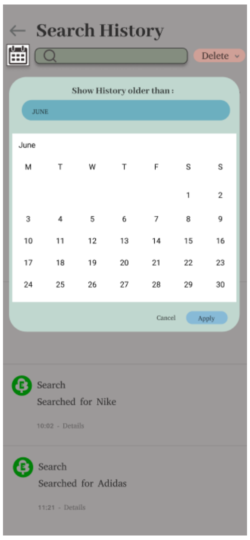
User 1: https://youtu.be/iwLvAVtlquI
User 2: https://youtu.be/CehlP1ieec8
User 3: https://youtu.be/1TnDr_X-FQM

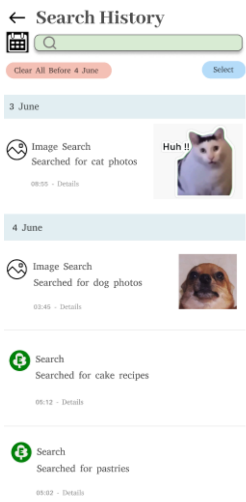
Task 3: Use history searching feature to check browsed content on 4th June by using calendar feature and **clear** the history of 3rd June.

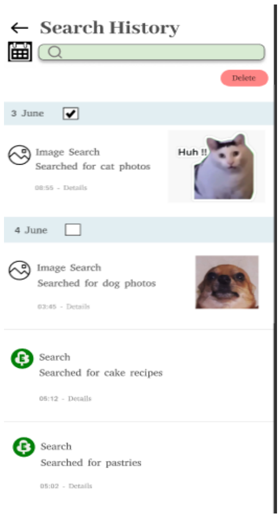


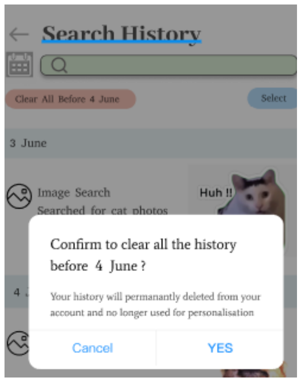


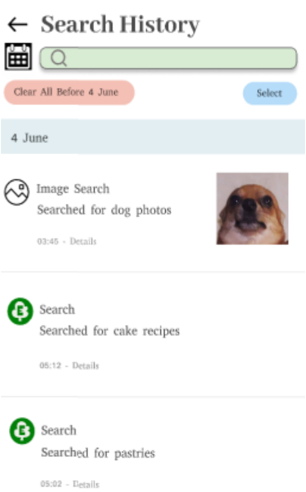












User 1: https://youtu.be/tfdrKmePjPI
User 2: https://youtu.be/4kOvTf8LQiU
User 3:https://youtu.be/KkXXhQoZcdc

Observations

Task 1: Use the **satellite** layer to find directions to TRX (drive)

User 1:

1. User 1 read the index card and understand it
2. The user click on the search bar.
3. The user input “TRX” through keyboard.
4. User click on the “Map” button.
5. User click transport button.
6. User tab “GO” button.
7. User choose type of map and click on the Satellite layer
8. User finished his task.

User 2:

1. User 2 understand the task.
2. The user click on search bar.
3. User type TRX.
4. User select the Map option.
5. User select the transport symbol.
6. User press layer symbol.
7. User choose a satellite layer for map view.
8. User finished her task

User 3:

1. User 3 reads the task and understands it.
2. The user click on the search bar.
3. User input TRX.
4. User click Magnifier icon to search.
5. User click on the Map option.
6. User press the “Transport button” .
7. User pick between the transports button and finally choose the Car option
8. User click the layer button.
9. User choose between 4 layers and finally choose the Satellite option
10. User finish the task.

<u>Summary of result interviews for task 1</u>

During the interview, User 1 mentioned difficulty in initiating directions after selecting map details. User 2 suggested an improvement: the current map lacks a zoom feature for detailed viewing, and she hopes this feature can be added. She also noted that the overall system is good, especially appreciating the smooth searching process. User 3 shared her opinion that the system is quite good due to its easy-to-understand icons and user-friendly design.

Task 2: Use image searching to capture an apple in **HD** and find the related content

User 1:

1. User 1 reads the task and understands it.

2. The user decides to capture an apple.
3. The user presses the camera icon at the base page.
4. The user clicks the camera icon to capture a photo of the apple.
5. The user changes the STD to HD before capturing the photo.
6. The user clicks the search icon.
7. The user has completed the task.

User 2:
1. User 2 understood the task clearly.
2. The user wants to capture an apple.
3. The user clicks the camera icon at the search bar from the main page.
4. The user presses the icon for search with the camera.
5. The user clicks the search icon to capture the photo of the apple.
6. The user thinks that the photo taken is not clear and the content is not related.
7. The user clicks the STD icon to the HD icon, the photo becomes clear and the content is related to the photo .
8. The user completes the task.

User 3:
1. User 3 receives the task and understands it.
2. The user clicks the camera icon at the base page.
3. The user presses the camera icon to capture a photo.
4. The user captures the photo of the apple by pressing the search icon.
5. The user uses STD to capture the photo and she realises that the content search is not related.
6. The user clicks the return icon.
7. The user changes the STD to HD and presses the search icon again.
8. The user thinks that the photo taken is more clear and the content is related to the photo.
9. The user has completed the task.

<u>Summary of result interviews for task 2</u>

During the interview, user 1 says that he did not face any problem when doing the task. User 2 mentions that the image searching feature is convenient to use. User 3 says that she did not face any difficulties when doing the task and the icons are clear and easy to use.

Task 3: Use history searching feature to check browsed content on 4th June by using calendar feature and **clear** the history of 3rd June.

User 1:
1. User 1 reads the task and understands it clearly.

2. The user opens up the Triple Bar.
3. The user presses the “History” option.
4. The user clicks on the Calendar icon.
5. The user selects the desired Month.
6. The user selects the specific Day.
7. The user clicks on the Select button.
8. The user selects the desired item to delete.
9. The user proceeds to click the Delete button.
10. User 1 accomplished his task.

User 2:

1. User 2 understands the task clearly.
2. The user clicks the Triple Bar.
3. The user selects the “History” option.
4. The user clicks on the Calendar icon beside the search bar of the Search History page.
5. The user presses the desired Month.
6. The user selects the specific Day.
7. The user clicks on the Select button.
8. The user selects the Date to delete.
9. The user clicks the Delete button.
10. User 2 accomplished her task.

User 3:

1. User 3 reads the task and understands it clearly.
2. The user clicks the Triple Bar.
3. The user selects the “History” option.
4. The user clicks on the Calendar icon.
5. The user presses the desired Month and specific Day
6. The user clicks on the Select button.
7. The user clicks the tickbox of the Date to be selected.
8. The user clicks the Delete button.
9. User 3 accomplished her task.

Summary of result interviews for task 3

During the interview, user 1 informed that there are no scrolling features available and states that it was a minor issue of inconvenience. User 2 mentions that there were no issues encountered and compliments on incorporating the calendar feature to find the related browsed content. User 3 says that it has smooth loading time and had good performance however the selected date chosen also displayed the dates before the selected date which was a minor issue to her and suggests that it should be more specified and not to show other dates.

Findings

After running all usability testing for our enhanced Ecosia search engine, several problems were identified across different tasks.In searching for location task, user 1 experienced difficulty to initiate directions after selecting map details. To solve this problem we will fix it by adding an obvious “Get Directions” button that shows up after selecting map details. Furthermore, it can also be improved by offering step-by-step instructions for starting the directions. Besides, user 2 pointed out that a zoom tool would be helpful for examining the map in detail. Thus,solutions could be verified by providing a pinch-to-zoom function on the map in addition to zoom-in and zoom-out for more detailed viewing.

For task 2 which is searching for images. There are no usability issues pointed out by users. All three users said that the search image features are convenient and user-friendly. This means that the current image searching functions and design should remain. Nevertheless, it is important to keep an eye out for any possible problems.

All users successfully completed task 3, which is searching for history, without any significant issues. Users have provided excellent feedback, with the calendar tool being particularly appreciated for its efficiency in rapidly accessing certain information. However, user 1 notices a lack of scrolling options in this part. To address this issue, we will add a scrolling option to the history searching interface to increase its usability. Furthermore, user 3 noted that the system would display dates before the set dates, which was confusing. We will change the calendar feature to display only the history of the selected date, which will enhance accuracy and the overall user experience.

In conclusion, the proposed solutions for the map interface will address the issues by adding a clear “Get Directions” button that appears after selecting map details and implementing a pinch-to-zoom function. Additionally, we will keep the current image search feature design while continuing to monitor for any potential issues that may occur in the future. Lastly, we will enhance the history interface with a scrolling feature and modify the calendar settings to ensure the system is efficient and user-friendly.